Homework 8
Elementary Linear Algebra

(MATH 2250-03), Spring, 2018

Check: 04/19/2019

Section 5.1 - Eigenvectors and Eigenvalues

1. Is $\lambda = 2$ an eigenvalue of $\begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$? Why or why not?

2. Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$? If so, find the eigenvalue.

3. Let $A = \begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix}$. Find a basis for the eigenspace corresponding to the eigenvalue $\lambda = 3$.

4. Show that if A^2 is the zero matrix, then the only eigenvalue of A is 0.

Section 5.2 - The Characteristic Equation

5. Find the characteristic polynomial and the eigenvalues of the matrix $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$.

6. Let A be an $n \times n$ matrix and suppose A has n real eigenvalues, $\lambda_1, \ldots, \lambda_n$, repeated according to their multiplicities, so that

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\ldots(\lambda_n - \lambda).$$

Show that $\det(A)$ is equal to the product of the eigenvalues of A repeated according to their multiplicities, i.e. show that

$$\det(A) = \lambda_1 \lambda_2 \ldots \lambda_n.$$

Section 5.3 - Diagonalization

7. The matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ may be factored in the form $A = PDP^{-1}$ as

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 1/2 & -3/4 \\ 1/4 & -1/2 & 1/4 \end{bmatrix}.$$

Use the Diagonalization Theorem to find the eigenvalues of A and a basis for each eigenspace.

8. Suppose A has n linearly independent eigenvectors. Prove that A^T also has n linearly independent eigenvectors. *(HINT: Use the Diagonalization Theorem)*